

# BKIT Bhalki — Official VTU MCA Complete Syllabus & Course Structure

**Institution:** Bheemanna Khandre Institute of Technology (BKIT), Bhalki (Bidar District, Karnataka)  
**Affiliated University:** Visvesvaraya Technological University (VTU), Belagavi  
**Scheme:** VTU 2022 / 2024 Scheme (Choice Based Credit System - CBCS & Outcome Based Education - OBE)  
**Degree:** Master of Computer Applications (MCA) — 2 Years (4 Semesters)  
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## Syllabus Governance at BKIT Bhalki

**Bheemanna Khandre Institute of Technology (BKIT)** is an affiliated engineering college under **Visvesvaraya Technological University (VTU)**.

- Under VTU statutes, all affiliated institutions follow the **Centralized VTU MCA Curriculum & Examination Regulations**.
- The examination question papers are set directly by VTU Belagavi, and evaluations are moderated under the **VTU Regional Center, Kalaburagi**.
- **Total Program Credits: 80 Credits** across 4 Semesters (approx. 20 credits per semester).
- **Grading Pattern:** 10-Point CGPA scale.
- **Evaluation Pattern:** Split into **Continuous Internal Evaluation (CIE)** conducted at BKIT and **Semester End Examination (SEE)** conducted by VTU.

## Summary of 4-Semester Course Structure

|  SEMESTER 1  
*(Foundations)* |  SEMESTER 2  
*(Core Development)* |  SEMESTER 3  
*(Advanced & Electives)* |  SEMESTER 4  
*(Industry & Project)* | |---|---|---|---| | • **Math Foundations** ( 22MCA11 ) | • **DBMS & SQL** ( 22MCA21 ) | • **Python Analytics** ( 22MCA31 ) | • **Advanced Web/Cloud** ( 22MCA41 ) | • **Operating Systems** ( 22MCA12 ) | • **Java OOP** ( 22MCA22 ) | • **IoT Architecture** ( 22MCA32 ) | • **DevOps Elective** ( 22MCA42X ) | • **Data Structures** ( 22MCA13 ) | • **Software Engg.** ( 22MCA23 ) | • **Cloud Computing** (*Elective*) | • **Technical Seminar** ( 22MCA43 ) | • **Computer Networks** ( 22MCA14 ) | • **Web Technologies** ( 22MCA24 ) | • **Cybersecurity** (*Elective*) | • **Major Project** (12 Credits) | | • **Algorithms (DAA)** ( 22MCA15 ) | • **DBMS Lab** ( 22MCA125 ) | • **Python Data Lab** ( 22MCA135 ) | • **Mandatory**

**Internship** (16 Weeks) | | • **DSA / OS Labs** | • **Java & Web Labs** | • **Mini Project** (2 Credits) | (Done in Bangalore/Hyderabad) | | **Total: 20 Credits** | **Total: 20 Credits** | **Total: 20 Credits** | **Total: 20 Credits** |

## SEMESTER 1 — DETAILED SYLLABUS

### Course Matrix (Semester 1)

| Course Code  | Course Title                                      | Type | Lecture (L) | Tutorial (T) | Practical (P) | Credits   | CIE Marks  | SEE Marks  | Total      |  |
|--------------|---------------------------------------------------|------|-------------|--------------|---------------|-----------|------------|------------|------------|--|
| 22MCA11      | Mathematical Foundation for Computer Applications | PCC  | 3           | 0            | 0             | 3         | 50         | 50         | 100        |  |
| 22MCA12      | Operating System Concepts                         | PCC  | 3           | 0            | 0             | 3         | 50         | 50         | 100        |  |
| 22MCA13      | Data Structures with Algorithms                   | IPCC | 3           | 0            | 2             | 4         | 50         | 50         | 100        |  |
| 22MCA14      | Computer Networks                                 | PCC  | 3           | 0            | 0             | 3         | 50         | 50         | 100        |  |
| 22MCA15      | Design and Analysis of Algorithms                 | PCC  | 3           | 0            | 0             | 3         | 50         | 50         | 100        |  |
| 22MCAL16     | Data Structures Laboratory                        | PCCL | 0           | 0            | 3             | 1.5       | 50         | 50         | 100        |  |
| 22MCAL17     | OS and Networking Laboratory                      | PCCL | 0           | 0            | 3             | 1.5       | 50         | 50         | 100        |  |
| 22RMI18      | Research Methodology & IPR                        | AEC  | 1           | 0            | 0             | 1         | 50         | 50         | 100        |  |
| <b>Total</b> |                                                   |      | <b>16</b>   | <b>0</b>     | <b>8</b>      | <b>20</b> | <b>400</b> | <b>400</b> | <b>800</b> |  |

### Module-Wise Breakdown (Semester 1)

#### 1. 22MCA11: Mathematical Foundation for Computer Applications

- **Module 1 (Mathematical Logic):** Propositions, Logical Connectives, Truth Tables, Tautology, Contradiction, Logical Equivalence, De Morgan's Laws, Predicates, and Quantifiers ( $\forall$ ,  $\exists$ ).
- **Module 2 (Set Theory & Relations):** Sets, Power Sets, Cartesian Product, Relations, Equivalence Relations, Equivalence Classes, Posets, and Hasse Diagrams.
- **Module 3 (Functions & Matrices):** Injective, Surjective, Bijective functions, Inverse functions, Matrix operations, Determinants, Matrix Inversion, Systems of Linear Equations, Eigenvalues & Eigenvectors.
- **Module 4 (Graph Theory):** Basic definitions, Degree of vertices, Handshaking Lemma, Paths, Cycles, Bipartite Graphs, Eulerian & Hamiltonian Graphs, Trees, Adjacency & Incidence Matrices, Planar Graphs, Euler's Formula ( $V - E + F = 2$ ).
- **Module 5 (Probability & Statistics):** Sample spaces, Axioms of probability, Conditional probability, Law of Total Probability, Bayes' Theorem, Random variables, Mean, Variance, Binomial and Normal distributions.

#### 2. 22MCA12: Operating System Concepts

- **Module 1:** Introduction to Operating Systems, Dual-Mode Operations (Kernel vs User), System Calls, Process Concepts, Process States, PCB, Context Switching, Inter-Process Communication (IPC).
- **Module 2:** CPU Scheduling: Criteria, Preemptive vs Non-Preemptive, FCFS, SJF, SRTF, Round Robin (RR), Priority Scheduling, Multi-Level Queue Scheduling.
- **Module 3:** Process Synchronization: Race Conditions, Critical Section Problem, Peterson's Algorithm, Semaphores (Counting and Binary), Mutex Locks, Classic IPC Problems (Producer-Consumer, Dining Philosophers).
- **Module 4:** Deadlocks: 4 Coffman Conditions, Resource Allocation Graphs (RAG), Deadlock Prevention, Avoidance with Banker's Algorithm, Deadlock Detection and Recovery.

- **Module 5:** Memory Management: Contiguous Allocation, Paging, Page Tables, TLB Cache, Segmentation, Virtual Memory, Demand Paging, Page Fault Handling, Page Replacement Algorithms (FIFO, LRU, Optimal), Thrashing.

### 3. 22MCA13: Data Structures with Algorithms

- **Module 1:** Asymptotic Analysis (Big- $O$ ,  $\Omega$ ,  $\Theta$ ), Array operations, Stacks (Array and Pointer representations, Infix to Postfix conversion, Postfix evaluation).
- **Module 2:** Queues: Linear Queue, Circular Queue, Deque, Priority Queue.
- **Module 3:** Linked Lists: Singly Linked List, Doubly Linked List, Circular Linked List, Linked List reversal, Polynomial Addition.
- **Module 4:** Trees: Binary Trees, Binary Search Trees (BST), Tree Traversals (Inorder, Preorder, Postorder), AVL Trees (Rotations: LL, RR, LR, RL), B-Trees.
- **Module 5:** Graphs: Adjacency representations, BFS, DFS, Dijkstra's Shortest Path, Prim's and Kruskal's Minimum Spanning Tree (MST).

### 4. 22MCA14: Computer Networks

- **Module 1:** Physical Layer, Topologies (Star, Mesh, Ring, Bus), OSI 7-Layer Model vs TCP/IP Protocol Suite, Transmission Media.
- **Module 2:** Data Link Layer: Framing, Error Detection (Parity, Checksum, CRC Modulo-2 Division), Flow Control (Stop-and-Wait, Sliding Window, Go-Back-N, Selective Repeat), CSMA/CD.
- **Module 3:** Network Layer: IPv4 Addressing, Subnetting & CIDR calculations, IPv6, NAT, ARP, Routing Algorithms (Distance Vector Bellman-Ford, Link State Dijkstra).
- **Module 4:** Transport Layer: UDP vs TCP, TCP 3-Way Handshake, TCP 4-Way Teardown, Flow Control, Congestion Control (Slow Start, AIMD).
- **Module 5:** Application Layer: DNS, HTTP/HTTPS, SMTP, Firewalls (Packet Filtering, Stateful Inspection), SSL/TLS Cryptographic Handshake.

### 5. 22MCAL16 & 22MCAL17: Practical Laboratories

- **Lab 1 (DSA):** C/C++ implementations of Stacks, Queues, Circular Queues, Linked Lists, BST insertion/traversal, AVL tree rotations, Dijkstra's algorithm.
- **Lab 2 (OS & Networking):** Linux bash shell scripting, Process creation using `fork()`, CPU scheduling simulation in C (FCFS/SJF/RR), Banker's algorithm implementation, Socket programming in C (Client-Server TCP/UDP).

## SEMESTER 2 — DETAILED SYLLABUS

### Course Matrix (Semester 2)

| Course Code | Course Title                               | Type | Lecture (L) | Tutorial (T) | Practical (P) | Credits | CIE Marks | SEE Marks | Total |
|-------------|--------------------------------------------|------|-------------|--------------|---------------|---------|-----------|-----------|-------|
| 22MCA21     | Database Management System (DBMS)          | IPCC | 3           | 0            | 2             | 4       | 50        | 50        | 100   |
| 22MCA22     | Object Oriented Programming Using Java     | IPCC | 3           | 0            | 2             | 4       | 50        | 50        | 100   |
| 22MCA23     | Software Engineering & Agile Methodologies | PCC  | 3           | 0            | 0             | 3       | 50        | 50        | 100   |
| 22MCA24     | Web                                        |      |             |              |               |         |           |           |       |

**Technologies** | PCC | 3 | 0 | 0 | 3 | 50 | 50 | 100 | | **22MCAL25| DBMS Laboratory** | PCCL | 0 | 0 | 3 | 1.5 | 50 | 50 | 100 | | **22MCAL26| Java Programming Laboratory** | PCCL | 0 | 0 | 3 | 1.5 | 50 | 50 | 100 | | **22MCAL27| Web Technologies Laboratory** | PCCL | 0 | 0 | 3 | 1.5 | 50 | 50 | 100 | | **22AEC28 | Ability Enhancement Course (Technical Communication)** | AEC | 1 | 0 | 0 | 1.5 | 50 | 50 | 100 | | **Total** | | **13** | **0** | **13** | **20** | **400** | **400** | **800** |

## Key Highlights (Semester 2)

- **22MCA21 (DBMS):** Relational Algebra, SQL (DDL, DML, TCL, Joins, Triggers, Stored Procedures), Functional Dependencies, Normal Forms (**1NF, 2NF, 3NF, BCNF**), ACID Transactions, Serializability, Two-Phase Locking (2PL), Crash Recovery (WAL, Checkpoints).
- **22MCA22 (Java):** JVM Architecture, Bytecode, Encapsulation, Inheritance, Dynamic Method Dispatch (Runtime Polymorphism), Abstract Classes vs Interfaces, Custom Exceptions, Multithreading & Synchronization, Collections Framework ( `ArrayList` , `HashMap` , `TreeSet` ).
- **22MCA23 (Software Engineering):** Software Process Models (Waterfall, Spiral, Agile Scrum, Kanban), Requirement Engineering, UML Diagrams (Use Case, Class, Sequence), Software Testing (Black-Box vs White-Box).
- **22MCA24 (Web Technologies):** HTML5 Semantic Elements, CSS3 Flexbox & Grid, JavaScript ES6+ (Arrow Functions, Closures, Promises, `async/await` ), DOM Manipulation, Form Validation with Regular Expressions, REST APIs.

## SEMESTER 3 — ADVANCED SPECIALIZATION & ELECTIVES

### Course Matrix (Semester 3)

| Course Code | Course Title | Type | Credits | |---|---|---|:---:| | **22MCA31 | Data Analytics Using Python** | IPCC | 4 | | **22MCA32 | Internet of Things (IoT)** | PCC | 3 | | **22MCA33X| Professional Elective 1 (Cloud Computing & Architecture)** | PEC | 3 | | **22MCA34X| Professional Elective 2 (Cybersecurity & Ethical Hacking)** | PEC | 3 | | **22MCAL35| Python Data Analytics Laboratory** | PCCL | 1.5 | | **22MCAL36| Mini Project** (Full-Stack / Cloud / Cyber application) | Project | 2 | | **22AEC37 | Societal Project / Community Internship** | Activity | 3.5 | | **Total** | | | 20 |

### Electives to Choose for Your Career Path (DevSecOps Track):

## SEMESTER 4 — INDUSTRY INTERNSHIP & MAJOR DISSERTATION

### Course Matrix (Semester 4)

| Course Code     | Course Title                                                         | Type       | Credits   | CIE Marks  | SEE Marks  | Total      | --- | --- | --- | --- | --- | --- | --- | --- |
|-----------------|----------------------------------------------------------------------|------------|-----------|------------|------------|------------|-----|-----|-----|-----|-----|-----|-----|-----|
| <b>22MCA41</b>  | <b>Advances in Web Technologies / Cloud Native Microservices</b>     | IPCC       | <b>3</b>  | 50         | 50         | 100        |     |     |     |     |     |     |     |     |
| <b>22MCA42X</b> | <b>Professional Elective 3 (DevOps &amp; Infrastructure as Code)</b> | PEC        | <b>3</b>  | 50         | 50         | 100        |     |     |     |     |     |     |     |     |
| <b>22MCA43</b>  | <b>Technical Seminar</b> (Research Paper Presentation)               | Activity   | <b>2</b>  | 100        | -          | 100        |     |     |     |     |     |     |     |     |
| <b>22MCA44</b>  | <b>Major Project Work / Industry Dissertation</b>                    | Project    | <b>9</b>  | 100        | 100        | 200        |     |     |     |     |     |     |     |     |
| <b>22MCA45</b>  | <b>Mandatory Industry Internship (16 Weeks)</b>                      | Internship | <b>3</b>  | 50         | 50         | 100        |     |     |     |     |     |     |     |     |
| <b>Total</b>    |                                                                      |            | <b>20</b> | <b>350</b> | <b>250</b> | <b>600</b> |     |     |     |     |     |     |     |     |

## How Your BKIT MCA Syllabus Directly Builds Your DevSecOps Career

| Semester | Subject in BKIT Syllabus | Skill It Teaches | DevSecOps Industry Relevance | |:---:|---|---|---| | **Sem 1** | Operating System Concepts | Linux Kernel, Process Scheduling, Threads, Memory | **Linux System Administration** for Cloud Servers | | **Sem 1** | Computer Networks | IPv4/IPv6, Subnetting, TCP 3-Way Handshake, Firewalls | **Cloud Virtual Private Clouds (VPC) & Security Groups** | | **Sem 1** | Mathematical Foundations | Logic, Graph Theory, Bayes' Probability | **Algorithmic Security & Threat Risk Modeling** | | **Sem 2** | Database Management Systems | SQL, Triggers, Indexing, Transactions | **Managing Cloud Relational Databases (AWS RDS, PostgreSQL)** | | **Sem 2** | Java & Web Technologies | Backend APIs, Exception Handling, HTTP REST | **Understanding Application Vulnerabilities & Microservices** | | **Sem 3** | Cloud Computing (Elective 1) | AWS EC2, S3, IAM, Virtualization | **Cloud Engineering & Infrastructure Architecture** | | **Sem 3** | Cybersecurity (Elective 2) | RSA Cryptography, OWASP Top 10, IDS/IPS | **Application & Cloud Security (AppSec / SecOps)** | | **Sem 4** | DevOps & Cloud Native (Elective 3) | Docker, Kubernetes, CI/CD, Terraform | **Full DevSecOps Pipeline Automation** | | **Sem 4** | Major Project & Internship | 6-Month Real Industry Project | **Securing a ₹10L–₹18L Full-Time Job Offer!** |

## Official Verification Links

### 1. VTU Official PG Scheme & Syllabus Portal:

 <https://vtu.ac.in/en/pg-scheme-syllabus/>

### 2. BKIT Bhalki Official Department Page:

 <http://recbhalki.org> → Academics → Department of MCA

### 3. Your Downloaded Question Banks with Complete Answer Keys:

 AI-Knowledge-Base/  FREE BOOKS/Question-Banks-PDFs/

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